Collaborative Document Day 1 RSST

2026-09-15-research-software-support-training

Welcome to The Workshop Collaborative Document.

This Document is synchronized as you type, so that everyone viewing this page sees the same text. This allows you to collaborate seamlessly on documents.

This is the Document for today: https://edu.nl/nmwaj

Collaborative Document day 1: https://edu.nl/nmwaj

Collaborative Document day 2: https://edu.nl/jnqpv

Collaborative Document day 3: https://edu.nl/g6eqm

Collaborative Document day 4: https://edu.nl/mwc48

Collaborative Document day 5: https://edu.nl/6jm6x

Code of Conduct

Participants are expected to follow these guidelines:

- Use welcoming and inclusive language.
- Be respectful of different viewpoints and experiences.
- Gracefully accept constructive criticism.
- Focus on what is best for the community.
- Show courtesy and respect towards other community members.

If you feel that the code of conduct is breached, please talk to one of the instructors (if the complaint is for one of the participants) or send an email to training@esciencecenter.nl (if the complaint is for one of the instructors).

Certificate of attendance

If you attend the full workshop we will send you a certificate of attendance.
We will delete all personal identifyable information 8 months after the training.

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Getting help

To ask a question, raise your hand in zoom. Click on the icon labeled "Reactions" in the toolbar on the bottom center of your screen,
then click the button 'Raise Hand '. For urgent questions, just unmute and speak up!

You can also ask questions or type 'I need help' in the chat window and helpers will try to help you.
Please note it is not necessary to monitor the chat - the helpers will make sure that relevant questions are addressed in a plenary way.
(By the way, off-topic questions will still be answered in the chat)

Workshop website & Lesson materials

RSST Sept-Oct 2026 (https://esciencecenter-digital-skills.github.io/2026-09-15-ds-rss-masterclass/)

RSST lesson material (https://esciencecenter-digital-skills.github.io/research-software-support/)

Instructors

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Helpers

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Agenda

Time	Topic
9:00	Welcome and icebreaker
9:30	Research Software
10:30	Coffee break
10:45	Software in the Research Landscape
11:45	Break
12:00	Landscape exercises
13:00	End of Day 1

Exercises

1. What is Research Software?

1.1 Is it Research Software

Results from the quiz (https://github.com/esciencecenter-digital-skills/2026-09-15-ds-rss-masterclass/blob/main/files/is_it_research_software_quiz.pdf)

1.2 Research Life Cycle

Research Life Cycle exercise (https://esciencecenter-digital-skills.github.io/research-software-support/modules/researchsoftware/exercise-research-life-cycle)

Discuss these two questions in your breakout room:

- Would you consider this Research Software or Software used in Research? Explain why.
- Where does it fit in the research life cycle? Explain why.

Group 4 - WORCS step-by-step procedure that researchers can follow to make a research project open and reproducible

- Would you consider this Research Software or Software used in Research? Explain why. Tool itself is Research Software as it is open source and developed for research purposes.
- Where does it fit in the research life cycle? Explain why: We think it belongs to the stage helping navigate the entire research workflow up until the end of the project. The purpose is to make the research reproducible, including the Publishing and Preserving

:mountain: 2. Research Software Landscape

2.1 Study and Present

Study and Present exercise (https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwarelandscape/exercise1)

You have 15 min for this exercise.
Please prepare to report back and summarize your findings.

Group 1. The Turing Way

- https://book.the-turing-way.org/
- What kind of "thing" did you choose? (e.g. an organization, group of people, document, set of guidelines, etc.) - the book, community and globar collaboration
- What is the core purpose or central thought behind your chosen concept? - to provide good research practices, resources and build community to advance best practices
- Is your chosen concept globally active, bound to a country or somehow geographically located somewhere
- Is your chosen concept globally active, bound to a country or somehow geographically located somewhere - global
- Describe the timeline of your chosen concept: any relevant history, its inception, any updates, is it still active? started 2019 and is still active (community calss are there)
- How can the scientific community benefit from this concept? - have guidlines and practical recommendations for reprodicible research,collaboration (explain GIT)).
- Are there other concepts related to your chosen concept? Version Control for instance is commonly used by user communities
- Fun fact: Esther Plomp has been an eScience center fellow. Book dashes are held in NL as well.

Group 2: The Carpentries

- https://carpentries.org/
- Training in data, software and library carpentry skills.
- Train the trainer program to help people teach these skills.
- Carpentries lessons are happening around the country.

- TDCC-NES project called DC4NES: open call for people who want to become instructurs: https://community.data.4tu.nl/open-call-individual-applications-for-carpentries-instructor-training-seats/
- Fun fact: Eva and Fenne are both trainer for the carpentries.
- Toby Hodges (director of curriculum) will come to NL and give a talk on Thursday 19th of November (save the date!)

Group 3: Open Science NL

- Check out their calls! https://www.openscience.nl/calls
- https://opensciencefestival.nl/ on the 14th of October. Sold out unfortunately.
- Part of NWO, has a budget of 10-20 millon per year for 10 years.

Group 4: Software heritage

- https://www.softwareheritage.org/
- Preserves software in a way that it ensures its reusibility
- DANS and NWO also support this initiative

Group 5: The research software alliance

- https://www.researchsoft.org/
- A lot of international members, including the eScience center
- Eva and Fenne went to the conference last week in Sheffield
- Community calls open for anyone

2.2 Mapping the FAIR principles

[Mapping the FAIR principles exercise (https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwarelandscape/exercise2)](https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwarelandscape/exercise2)

PureGoME

- PureGoMe can be downloaded from the project repository, while metadata is accessible independently from the registry. -> F A
- PuReGoMe's GitHub repository has detailed records of the development history https://github.com/puregome . -> R
- The code includes dependencies to other software, such as various Python libraries. -> R
- PuReGoMe has a (versioned) DOI from Zenodo. -> F
- PuReGoMe uses the Apache 2.0 open source licence. ->
- PuReGoMe uses standard file formats (e.g., CSV files) for data exchange. -> I
- PuReGoMe refers to other objects such as websites.
- PureGoMe is registered in the Research Software Directory that captures the most relevant metadata, including the identifier, in searchable and indexable form. -> F

:book: Homework

- Check if your FAIR mapping was correct: https://www.nature.com/articles/s41597-022-01710-x

Collaborative Notes

- After finishing this training you will get a certificate
- Can miss 10% of the training
- Training materials are open and reusable
- We use the green check (Zoom > bottom of the screen > Reactions > green tick mark) for excercises

1. What Is Research Software?

1.1 What is Software?

Some history of software. Organ books and punch cards.

If anyone wants a deepdive into the punch cards/organ sheet, this video has a ton of interesting info on it https://youtu.be/jjNca1L6CUk (from the perspective of audioanimatronics developed at Disneyland in the early 1900s)

Software today? In which ways do you think it's different from the past?

- Faster
- Digital mostly nowadays
- More computational power
- Buy software or borrow from a CD; no open source available; required software key
- We write text rather than numbers as instructions

Software is data

- Can be stored as bits
- Read, loaded and processed
- Input and/or output
- Can be assigned a DOI, licence etc.

It's complex: it is creatively generated, interconnected and multi-layered.

It's interdependent: it builds upon and therefore depends on other software (and data).

It's executable: it needs to be run to have a value (to process data). A set of instructions needs to be executed, otherwise its just text. Software can only be interpreted by running it and seeing what it actually does.

It's fragile: the (technical) surroundings are dynamic, it can (and will) break soon and therefore needs to be regularly updated.

Software requires dedicated maintenance

Software needs to be actively maintained to remain useable, but maintenance is often lacking in academic contexts:

- lack of funding
- lack of incentives/rewards
- lack of understanding and expertise (maintenance is different from creation)

- individualistic work with fast turnover (think: developed by PhD/postdoc - after finishing the paper people move on, leaving a knowledge gap in the group)

Software Vs Data management

- Software is a living thing
- Good data management will not ensure good software management
- While some of the FAIR data management practices apply directly, others do not apply as easily

Take Home Messages

- Software are the instructions that tell hardware (computer) what to do
- Software is a "living thing"

1.2 Research Software

A photo of a blackhole, isn't just a picture. The instructions you give to the hardware determines how to convert an image.

Why Treat Research Software Differently?

- Acknowledge its importance during research
- Designates it as a research output; citeable, true deliverable of a grant application for example
- Enables proper attribution; DOI, publications
- Facilitates findability and reusability

What is Research Software?
Research software includes source code files, algorithms, scripts, computational workflows and executables that were created during the research process or for a research purpose. Software components (e.g., operating systems, libraries, dependencies, packages, scripts, etc.) that are used for research but were not created during or with a clear research intent should be considered software in research and not Research Software.

- Using powerpoint is not necessarily Research Software
- Using R is not necessarily Research Software, but the code/syntax you produce is research output

The survey tool mentioned was LimeSurvey (https://github.com/LimeSurvey)

The Research Lifecycle

Research Software is mainly used in collecting and processing & analyzing steps.

Take Home Messages

- Software is an important part of research
- Not all software used in research is Research Software
- Defining what it is provides recognition in the research community
- It's a moving target; research software evolves; it's not about giving a precise label, but about giving credit

:mountain: 2. Research Software Landscape

Take home messages

- Software is an important part of research
- Some problems arise due to the combination of:
    - the role of software in research
    - the nature of software
    - how software is developed
- Many initiatives have started in the last decade to tackle these issues
- The landscape is still under heavy development

Resources

- Open call for Carpentries Instructors: https://community.data.4tu.nl/open-call-individual-applications-for-carpentries-instructor-training-seats/
- LimeSurvey: https://github.com/LimeSurvey
- Workflow for Open Reproducible Code in Science (WORCS): https://cjvanlissa.github.io/worcs/index.html
- Open Science calls https://www.openscience.nl/calls
- Open Science festival https://opensciencefestival.nl/
- Coding cafes https://code-cafes-nl.github.io/cafe_playbook/
- List of events from Eva: https://researchsoftwaretraining.nl/%F0%9F%8D%82fall-training-events/

:car: Parking lot

:book: Homework

- Check if your FAIR mapping was correct: https://www.nature.com/articles/s41597-022-01710-x
- Optional: play around with the coding tool or share with collegues: https://www.online-python.com/
- Optional: further reading materials https://esciencecenter-digital-skills.github.io/research-software-support/modules/softwarelandscape/reading